MISSION 5

Problems :

> Problem 1 :
> Ilya said our D4 bifurcation diagram is correct.
> Actually , many years ago , somebody also solved the D4 bifurcation diagram ,
> but he solved it by hand , no computer software , no deepseek at that time.
> He has a paper on the internet (but Ilay don't know his name)
> The following is what we need to do :
> Find out this paper on the internet and learn the skills and technque of
> finding D4 bifurcation diagram.

> Problem 2 :
> After finished Problem 1 , try using the skills and technque you learned
> on that paper and generalize to our system :
> Hamiltonian system :
>
> $$\dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z}$$
>
> with$B_1 = 1, B_2 = 0$
> And to solved the bifurcation diagram $\Sigma := \Sigma_1 \cup \Sigma_2$

Trick :

> Theorem :
> If we suppose our $F(x, y, \lambda)$ satisfies $deg(F) = 3$
> Then we have :
>
> $$\lambda \in \Sigma_2 \iff F(x, y, \lambda) = Q_1 Q_2 + \alpha$$
>
> where
>
> $$Q_1 \text{ is polynomial with } deg(Q_1) = 1$$
>
> $$Q_2 \text{ is polynomial with } deg(Q_2) = 2$$
>
> $$\alpha := F(x_1, y_1) = F(x_2, y_2) \text{ with } (x_1, y_1), (x_2, y_2) \text{ balanced points}$$